An alternating Turán inequality for the Euler numbers

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MathDB 338918. Full proof of the assertion in Conjecture 3.3 of Sokal's accepted manuscript [1]. The source audit uses the complete author accepted version deposited at UCL; the separately typeset journal version remains unread. Review is by the authoring agent only.

Task 1: the full inequality

Define the Euler zigzag numbers by

$$\sec t + \tan t = \sum_{n\ge 0} E_n \frac{t^n}{n!}, \qquad e_n = E_n/n!.$$

Thus $E_0, E_1, E_2, E_3, E_4 = 1, 1, 1, 2, 5$; these are the positive secant and tangent numbers interleaved.

Theorem 1. *For all integers $n \ge 0$ and $j, k \ge 1$,*

$$(-1)^{n-1}\big(e_n e_{n+j+k} - e_{n+j}e_{n+k}\big) > 0. \tag{1}$$

Put $c = 4/\pi$, $a = 2/\pi$, and define

$$B(s) = \sum_{r\ge 0} \frac{(-1)^r}{(2r+1)^s} \quad (s \ge 1), \qquad L(s) = \sum_{r\ge 0} \frac{1}{(2r+1)^s} \quad (s > 1).$$

The classical partial fraction expansions of secant and tangent give

$$e_n = ca^n \begin{cases} B(n+1), & n \text{ even}, \\ L(n+1), & n \text{ odd}. \end{cases} \tag{2}$$

These formulas, including the conditionally convergent case $B(1) = \pi/4$, are recorded in [1, eqs. (3.3)–(3.5), (3.16)].

Lemma 1. *For $s \ge 1$, $0 < B(s) < 1$ and $B(s+2) > B(s)$. For $s > 1$, $L(s) > 1$ and $L(s+2) < L(s)$.*

Proof. The bounds for B follow from the alternating series, and those for L from its positive terms. The decrease of L is termwise, strictly at $r = 1$. For B subtract convergent partial sums to obtain

$$B(s+2) - B(s) = \sum_{r\ge 1} (-1)^{r+1} h_s(2r+1), \qquad h_s(x) = x^{-s} - x^{-s-2}.$$

For $x \ge 3$, $h_s(x) > 0$, $h_s(x) \to 0$, and

$$h_s'(x) = x^{-s-3}\big(s + 2 - sx^2\big) < 0.$$

Pairing consecutive terms proves that this alternating sum is strictly positive. No differentiation of an infinite series is used at $s = 1$. □

We recall with a proof the two known same-parity properties used below. They are established in [1, Section 3, Proposition 3.2].

Lemma 2. *The positive sequence $(e_{2m})_{m\geq 0}$ is strictly log-concave; $(e_{2m+1})_{m\geq 0}$ is strictly log-convex. Consequently, for integers $m \geq 0$ and $u, v \geq 1$,*

$$e_{2m}e_{2(m+u+v)} < e_{2(m+u)}e_{2(m+v)}, \qquad e_{2m+1}e_{2(m+u+v)+1} > e_{2(m+u)+1}e_{2(m+v)+1}.$$

Proof. Write $x_r = 4/[\pi^2(2r+1)^2]$. The cosine product yields

$$\sum_{m\geq 0} e_{2m}z^m = \prod_{r\geq 0}(1 - x_r z)^{-1}.$$

Here is an elementary strictness argument. For a sequence h set $T(h)_{ij} = h_{j-i}$, with $h_t = 0$ for $t < 0$. Every two-by-two minor of $T(1, x, x^2, \ldots)$ is nonnegative: if all four entries are nonzero its determinant is zero, and in the other cases it is either zero or the product of its diagonal entries. Convolution multiplies these upper triangular matrices. The two-by-two Cauchy–Binet formula therefore preserves nonnegative minors; each sum here is finite.

For the first two factors the coefficient is $h_m = \sum_{i=0}^{m} x_0^i x_1^{m-i}$, and direct algebra gives

$$h_m^2 - h_{m-1}h_{m+1} = (x_0x_1)^m > 0 \qquad (m \geq 1).$$

Multiply their Toeplitz matrix on the left by the matrices of the remaining factors. For rows $0, 1$ and columns $m, m+1$, the intermediate indices $0, 1$ in Cauchy–Binet contribute precisely the displayed positive minor: the left matrix has leading minor 1. All other contributions are nonnegative. The finite products thus have this same positive lower bound. Since $\sum x_r < \infty$, their coefficients converge to those of the infinite product, proving strict log-concavity of e_{2m}.

The tangent expansion gives $e_{2m+1} = 2\sum_{r\geq 0} x_r^{m+1}$. Strict Cauchy–Schwarz for this positive measure, supported at distinct positive points, proves strict log-convexity. All sums converge. Finally, strict log-concavity means that consecutive ratios strictly decrease. Multiplying u such ratios, starting at m and at $m+v$, proves the first rectangular inequality. Increasing ratios prove the second one. □

Proof of Theorem 1. In every product in (1), the positive factor c^2a^{2n+j+k} from (2) is the same, so it can be canceled.

If j, k are both even, apply Lemma 2 with $u = j/2$, $v = k/2$. If j, k are both odd and n is even, the first product after cancellation is a product of two B values, hence is less than 1; the second is a product of two L values, hence is greater than 1. The difference is negative. For odd n the two roles interchange, so it is positive.

In the remaining case, symmetry allows us to take j even and k odd. If n is even, the difference after cancellation is

$$B(n+1)L(n+j+k+1) - B(n+j+1)L(n+k+1) < 0.$$

Indeed, both factors of its first product are strictly smaller than the corresponding factors of its second, by Lemma 1 iterated $j/2$ times. If n is odd, the difference is

$$L(n+1)B(n+j+k+1) - L(n+j+1)B(n+k+1) > 0,$$

because both factors of its first product are strictly larger. All L arguments used are at least 2, all B arguments at least 1. This proves every case, including $n = 0$. □

Attribution and verification. The partial fraction expansions and the two same-parity properties are prior results, explicitly attributed above. The contribution is the complete parity argument for the conjectured three-parameter inequality. The accompanying exact verifier checks Euler coefficients by independent recurrences, finite instances of all parity classes, the product-minor identity, and the alternating-series comparison. Finite checks supplement the proof; they do not prove its unbounded quantifiers.

Four proposed follow-up tasks

These directions are not claimed as solved.

Task 2: For $b_n(q) = \sum_{r\geq 0}(-1)^{r(n+1)}(q+2r)^{-n-1}$, determine all $q > 0$ for which the analogue of (1) holds for every n, j, k.

Task 3: Determine the signs of the consecutive Hankel determinants $\det(e_{n+i+j})_{0\leq i,j<r}$ for every $n \geq 0$ and $r \geq 3$.

Task 4: For fixed j, k, find the exact leading asymptotic of $(-1)^{n-1}(e_n e_{n+j+k} - e_{n+j}e_{n+k})$ separately on even and odd n, with an explicit error bound.

Task 5: Construct a combinatorial proof of (1) after multiplication by $n!(n+j)!(n+k)!(n+j+k)!$, using alternating permutations.

References

[1] A. D. Sokal, *The Euler and Springer numbers as moment sequences*, arXiv:1804.04498v3 (2018), Conjecture 3.3, p. 14. Published in *Expositiones Mathematicae* **38** (2020), 1–26, doi:10.1016/j.exmath.2018.08.001. All 30 pages of the current manuscript were read. The UCL deposit explicitly identifies the same file as the accepted version: `https://discovery.ucl.ac.uk/id/eprint/10059424/`. The separately typeset journal version remains unread.